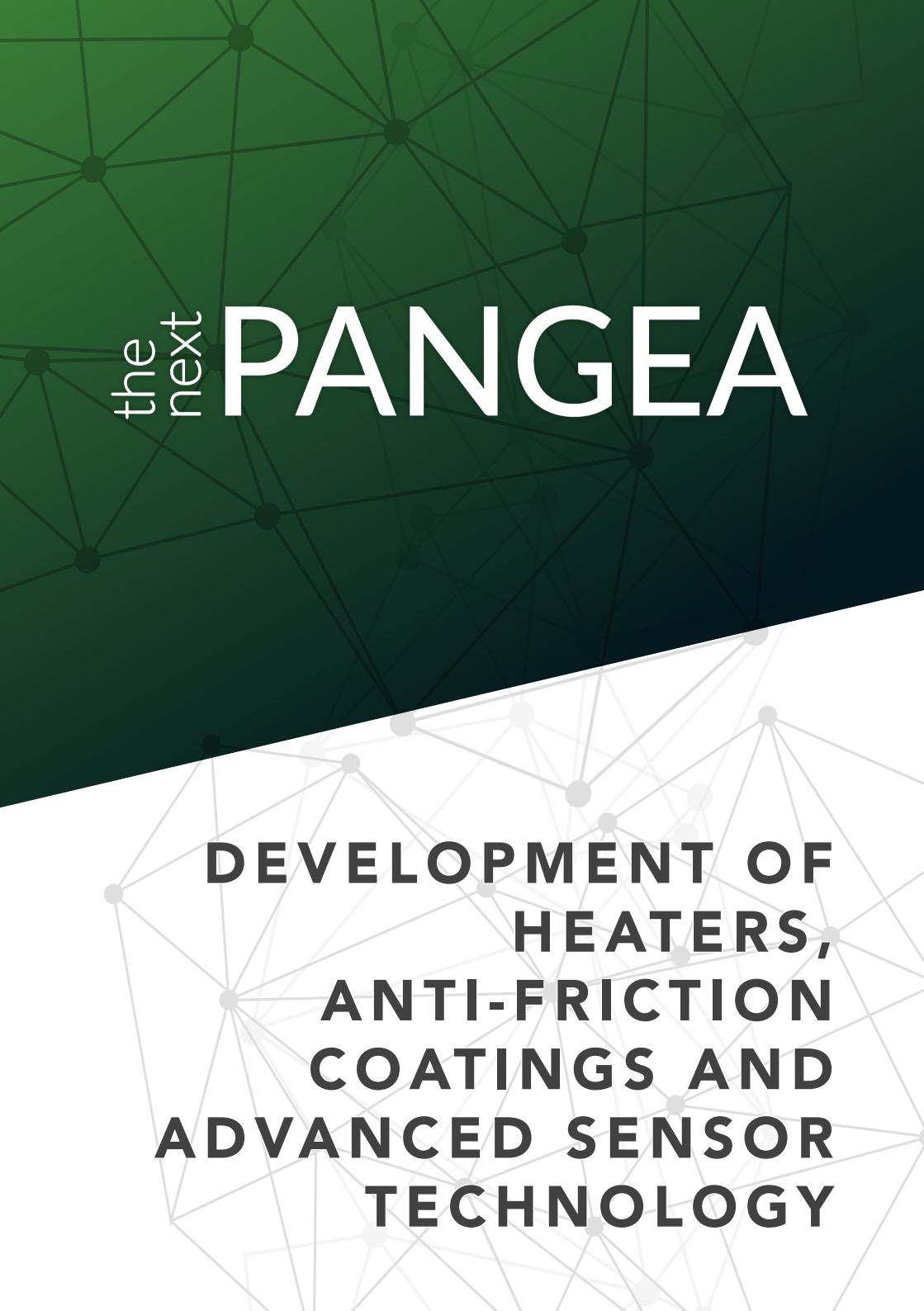




the next PANGEA

**DEVELOPMENT OF
HEATERS,
ANTI-FRICTION
COATINGS AND
ADVANCED SENSOR
TECHNOLOGY**

HEATERS

HIGH TECH HEAT BLANKETS

Development of adaptable heaters formulated with latest generation materials

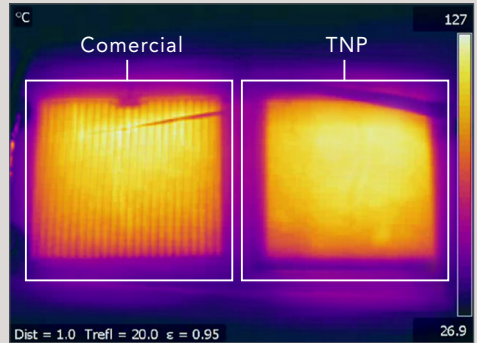
Flexible heating elements based on carbonaceous materials **for general or localized zone heating, with the possibility of manufacturing in custom geometries.** Designs are specifically tailored for repair zones of composite materials or high-temperature applications requiring adhesive curing at elevated temperatures.

TNP can develop heaters for application on flat or curved elements, as well as variable temperature management systems to adapt to various sectors such as aviation, vehicles, industrial products, and household appliances.

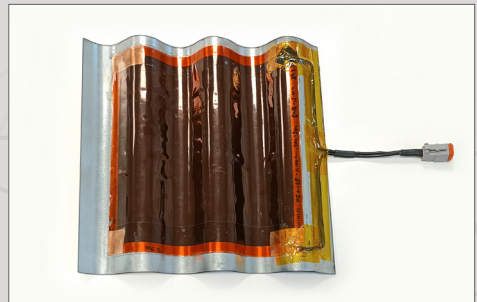
TNP's designs achieve improvements compared to other commercial products in the following cases:

Customized geometry

Development of heated tracks for custom complex geometries, ranging from small sizes to large surfaces, allowing a **perfect fit to the repair area** and the **creation of heaters with highly versatile complex and irregular shapes.**



Comparison of homogeneity between commercial blanket (left) and TNP design (right) of 200x200mm



TNP heater installed on a wavy surface to heat irregular material 300x300mm

Homogeneity

Achieving maximum thermal homogeneity of the assembly through the adjustment of heating bands and the combination of different thermal elements. Various comparisons with commercial blankets demonstrate superior homogeneity of the TNP model, providing a **consistent high-temperature layer without stripes or noticeable thermal differences** in the range of 40 to 150°C.

Compatibility

Compatibility with existing electrical power systems and thermal control, as well as the development of custom electrical controllers as needed.

ANTI FRICTION

SMART INKS

Coatings with nanomaterial-based compounds to combat friction and enhance mechanical performance

TNP has the capability to develop specific compounds to enhance performance in mechanical assemblies where friction and wear phenomena occur. As a result, **various formulations have been successfully developed, easily applicable in sprayable format or previously deposited on liners**, capable of substantially **improving the performance and lifespan** of these assemblies.

The different formulations have been designed to combat wear and reduce the coefficient of friction (COF) using advanced technology to optimize the nature and quantity of the formulated additives. The applied development yields final results on mechanical assemblies with the following key outcomes:

Coefficient of Friction

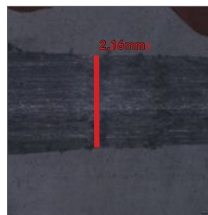
85% Reduction in Coefficient of Friction:

The treated parts exhibit an 85% decrease in the coefficient of friction compared to untreated trials, dropping from 0.77 to 0.15, leading to a significant increase in the lifespan of the assemblies.

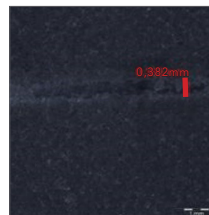


Sample of multiple components with potential application of Anti-Friction coatings

Wear footprint
Without Treatment:
2mm



Wear footprint
With TNP Treatment:
0.3mm



Wear footprint
10X Reduction
With TNP Treatment

Wear Footprint Comparison

Wear footprint

The wear footprint is considerably improved, with a 10-fold reduction (x10) observed. Parts without treatment had a wear footprint width of 2mm, whereas TNP treatment reduced it to just 0.3mm.

Perfilometry

The depth of the worn footprint is reduced from 170/180µm in untreated parts to 10/16µm with treatment using our technology.

SENSOR TECH

SCREEN PRINTING SENSORS

Development of sensors for deformations and cracks formulated with latest generation materials

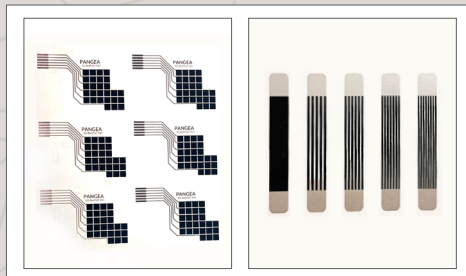
TNP's flexible graphene sensors, designed to measure stresses on surfaces, **enable the monitoring of structures, components, and areas with custom geometry and properties.** Thanks to their design, local measurements are transformed into a globally visualized metric across the entire surface where the sensor is applied.

Primarily targeted for metallic and composite surfaces, these sensors offer versatility in various environments due to their layered design and local sensitivities, making them comparable to strain gauges. However, they surpass strain gauges by covering much larger surfaces.

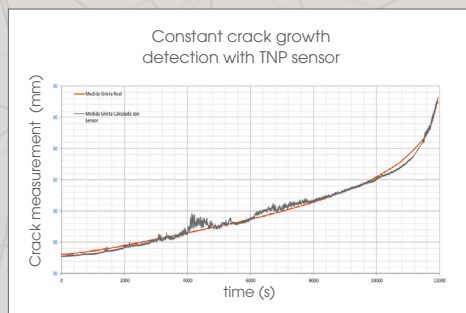
Mesh Sensor

The **mesh-shaped sensor design reduces the required number of cables and allows coverage of a much larger area compared to any strain gauge.**

Its flexibility and the number of sensors composing the mesh facilitate adaptation to complex geometries.



Samples of different screen-printed sensors



Graph of crack propagation

Adaptable Geometry

The adaptable geometries suit each project, enabling adjustments to new components or environments while maintaining low overall sensor thickness for maximum versatility.

High Sensitivity

These sensors can detect very small strain changes, at least 0.02%, and gauge factors (GF) surpass those of strain gauges, providing high sensitivity in detecting cracks and deformations.



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